

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	2026-09-28
Team ID	SWTID-2026-2362
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture Guidelines:

The solution architecture maps operational interfaces and logic gates across specific enterprise boundaries:

- **Infrastructural Demarcation:** Cloud SaaS Layer (ServiceNow Platform Core) interacting with Cloud PaaS Systems (Azure / Apple Enrollment Engines).
- **External Interfaces:** Multi-point REST integrations via IntegrationHub for live shipment tracking and corporate directories.
- **Data Storage:** Unified Configuration Management Database (CMDB) tracking live assets.

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Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	Self-Service workspace where employees browse catalog options, check hardware setups, and view ticket tracking lifecycles.	ServiceNow Employee Center, Service Portal widgets, HTML5, CSS3, AngularJS frameworks.

2.	Application Logic-1	Core intake rules, role filtering capabilities, form visibility constraints, and client-side data auto-population scripts.	ServiceNow Client Scripts, UI Policies, GlideSystem API arrays.
3.	Application Logic-2	Central process engine handling inventory stock checks, branching approvals, asset creation loops, and dynamic task generation.	ServiceNow Flow Designer low-code engine, Workflow core actions.
4.	Application Logic-3	Server-side execution layer processing complex database transactions, background evaluations, and operational scripts.	JavaScript, ServiceNow GlideRecord API routines, Business Rules.
5.	Database	Relational configuration engine capturing configuration data, active requests, fulfillment tasks, and application logs.	ServiceNow Native Relational Database Layer (MariaDB backend structure).
6.	Cloud Database	Cloud repository tracking real-time enterprise hardware definitions, global stocks, vendor models, and assignment states.	ServiceNow CMDB (Configuration Management Database) - alm_hardware tables.
7.	File Storage	System capturing transactional documents, digital invoice signatures, device handoff receipts, and configuration logs.	ServiceNow Attachment Repository Database layers (sys_attachment).
8.	External API-1	Endpoint fetching dynamic courier tracking statuses, delivery logs, and delivery	IntegrationHub REST API, Bluedart / DHL Webhook endpoints.

		events directly to request tickets.	
9.	External API-2	Cloud directory engine validating identity objects, team hierarchies, cost management profiles, and provisioning permissions.	Microsoft Azure AD Graph API / Okta SCIM connectors.
10.	Machine Learning Model	Smart forecasting model parsing historical consumption limits to alert teams before popular hardware configurations go out of stock.	ServiceNow Predictive Intelligence engine (Regression models).
11.	Infrastructure (Cloud)	Enterprise Software-as-a-Service architecture ensuring rapid platform scaling, managed servers, and automated infrastructure routines.	ServiceNow Cloud Infrastructure (Multi-instance secure cloud architecture).

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Script extensions, custom UI layouts, and asynchronous client logic tools processing responsive web modules inside portal dashboards.	JavaScript, AngularJS 1.x libraries, Bootstrap UI layouts.
2.	Security Implementations	Access rules checking validation roles, multi-factor login checks, identity	OAuth 2.0 validation tokens, Access Control Lists (ACLs), TLS 1.3 data transfer encryption

		checks, and encrypted field protocols.	protocols, Single Sign-On (SSO).
3.	Scalable Architecture	Enterprise-grade cloud model supporting simultaneous data queries across intake databases without altering core workflow logic.	Multi-Instance Architecture model, modular ServiceNow Application Scope boundaries.
4.	Availability	Globally redundant infrastructure ensuring hot-standby failover capabilities, load balancers, and real-time transaction mirroring layers.	High Availability (HA) node architecture configuration, F5 Networks Advanced Load Balancers.
5.	Performance	Performance design ensuring system response times under 2 seconds during stock check lookups, using indexed fields.	ServiceNow Cache Management architectures, Multi-thread Flow Engine execution arrays.